

fCC Task: *Workshop: Build a Report Card Printer*

Skills: *input(), float(), arithmetic, f-strings with :.1f*

Please use **Thonny** to complete the task. Add these lines at the top of your code:

```
print("Name: [Your Name]")
print("Class: [Your Class]")
print("Class Number: [Your Class Number]")
```

Instructions

Write a Python script that acts as a school report card printer. The script prompts the user for a student's name and three subject grades, calculates the average, and prints a formatted report card.

Key Rules & Concepts:

- ▶ `input()` always returns a string — use `float()` to convert it to a number before doing maths.
- ▶ f-strings: put `f` before the opening quote, use `{ }` to embed variables.
- ▶ `{average:.1f}` means: format average to exactly 1 decimal place.
- ▶ Print `'--- Report Card ---'` as a separator before the grades.

Objective: Follow each step below to complete the task.

Steps to Complete
1. Use <code>input()</code> to prompt for the student's name and store it in a variable named <code>name</code> .
2. Use <code>float(input())</code> to prompt for the English grade and store it in <code>english</code> .
3. Use <code>float(input())</code> to prompt for the Maths grade and store it in <code>maths</code> .
4. Use <code>float(input())</code> to prompt for the Science grade and store it in <code>science</code> .
5. Calculate the average: <code>average = (english + maths + science) / 3</code>
6. Print the separator string: <code>--- Report Card ---</code>
7. Use f-strings to print the student's name and each of the three grades.
8. Print the average using an f-string with <code>:.1f</code> to display exactly 1 decimal place.

Output Checklist (8 marks)
1. The script prompts for the student's name using <code>input()</code> .
2. The script prompts for the English grade and converts it with <code>float()</code> .
3. The script prompts for the Maths grade and converts it with <code>float()</code> .
4. The script prompts for the Science grade and converts it with <code>float()</code> .
5. The script calculates the average by summing all three grades and dividing by 3.

6. The script prints the separator '--- Report Card ---' on its own line.
7. The script prints the student's name, English, Maths, and Science using f-strings.
8. The average is printed with exactly 1 decimal place using the <code>:.1f</code> format specifier.

Submission (2 marks)

Please submit the following two files to Google Classroom:

- ①

A Python file named: `[YourClass][YourNumber].py` (e.g. `S3D05.py`)
- ②

A screenshot showing your complete code and output in the *Thonny IDE*.

Sample Run (inputs: Alice, 80, 90, 70)

```
Student name: Alice
English grade: 80
Maths grade: 90
Science grade: 70
--- Report Card ---
Student : Alice
English : 80.0
Maths   : 90.0
Science : 70.0
Average : 80.0
```

— — — End of Task — — —

fCC Task: *Workshop: Build an Employee Profile Generator*

Skills: *String concatenation (+), += operator, str() conversion, f-strings, string slicing*

Please use **Thonny** to complete the task. Add these lines at the top of your code:

```
print("Name: [Your Name]")
print("Class: [Your Class]")
print("Class Number: [Your Class Number]")
```

Instructions

Write a Python script that builds an employee profile using string operations. You will combine names with concatenation, append to an address using +=, convert numbers with str(), format a card with an f-string, and extract parts of a code string using slicing.

Key Rules & Concepts:

- ▶ You CANNOT concatenate a string and a number directly — use str() to convert first.
- ▶ string[start:stop] — the stop index is EXCLUSIVE (not included in the result).
- ▶ Negative index: string[-3:] means the last 3 characters.
- ▶ += is shorthand: address += ', 4B' is the same as address = address + ', 4B'

Objective: Follow each step below to complete the task.

Steps to Complete
1. Create variables first_name = 'John' and last_name = 'Doe'. Build full_name by concatenating them with a space in between.
2. Create address = '123 Main Street' then use += to append ', Apartment 4B'.
3. Create employee_age = 28. Build employee_info using str() to convert age, then print it (e.g. John Doe is 28 years old).
4. Create experience_years = 5. Build experience_info using str() conversion and print it.
5. Create position = 'Data Analyst' and salary = 75000. Build employee_card using an f-string that includes full_name, employee_age, position, and salary. Print it.
6. Create employee_code = 'DEV-2026-JD-001'. Use slicing [0:3] to get the department.
7. Use slicing [4:8] for the year code, [9:11] for the initials, and [-3:] for the last three digits.
8. Print all four sliced values on one line.

Output Checklist (8 marks)
1. full_name is built by concatenating first_name + ' ' + last_name.
2. address starts as '123 Main Street' and += appends ', Apartment 4B'.
3. str() is used to convert employee_age before concatenation in employee_info.

4. <code>str()</code> is used to convert <code>experience_years</code> in <code>experience_info</code> .
5. <code>employee_card</code> is built with an f-string showing name, age, position, and salary.
6. <code>employee_code[0:3]</code> correctly extracts the department tag 'DEV'.
7. <code>employee_code[4:8]</code> , <code>[9:11]</code> , and <code>[-3:]</code> extract the year, initials, and last digits.
8. All four sliced values are printed together on one line.

Submission (2 marks)

Please submit the following two files to Google Classroom:

- ① A Python file named: [YourClass][YourNumber].py (e.g. S3D05.py)
- ② A screenshot showing your complete code and output in the *Thonny IDE*.

Expected Output

```
John Doe is 28 years old
Experience: 5 years
Employee: John Doe | Age: 28 | Position: Data Analyst | Salary: $75000
DEV 2026 JD 001
```

— — — End of Task — — —

fCC Task: *Workshop: Build a Bill Splitter*

Skills: *Variables, float, += operator, multiplication, division, round()*

Please use **Thonny** to complete the task. Add these lines at the top of your code:

```
print("Name: [Your Name]")
print("Class: [Your Class]")
print("Class Number: [Your Class Number]")
```

Instructions

Write a Python script that calculates and splits a restaurant bill. The script uses a running total, adds all dish costs, calculates a 25% tip, and divides the final amount equally between friends.

Key Rules & Concepts:

- ▶ `round(number, 2)` rounds to 2 decimal places.
- ▶ 25% tip = multiply by 0.25 (or multiply by 25, then divide by 100).
- ▶ `running_total += x` adds x to `running_total` in place.
- ▶ Division (`/`) always returns a float in Python 3.
- ▶ Note: floating-point arithmetic may cause slight imprecision in intermediate values.

Objective: Follow each step below to complete the task.

Steps to Complete
1. Create <code>running_total = 0</code> and <code>num_of_friends = 4</code> .
2. Create the four cost variables: <code>appetizers = 37.89</code> , <code>main_courses = 57.34</code> , <code>desserts = 39.39</code> , <code>drinks = 64.21</code>
3. Add all four costs to <code>running_total</code> in one line using <code>+=</code> . Print 'Total bill so far:' followed by <code>running_total</code> .
4. Calculate <code>tip = running_total * 0.25</code> . Print 'Tip amount:' followed by <code>tip</code> .
5. Add the tip to <code>running_total</code> using <code>+=</code> . Print 'Total with tip:' and the new <code>running_total</code> .
6. Calculate <code>final_bill = running_total / num_of_friends</code> . Print 'Bill per person:' and <code>final_bill</code> .
7. Round the final bill: <code>each_pays = round(final_bill, 2)</code> .
8. Print 'Each person pays:' followed by <code>each_pays</code> .

Output Checklist (8 marks)
1. <code>running_total</code> is initialised to 0 and <code>num_of_friends</code> is set to 4.
2. All four cost variables are created with the correct values.
3. All costs are added to <code>running_total</code> in a single <code>+=</code> statement.

4. 'Total bill so far:' is printed with running_total.
5. tip is calculated as running_total * 0.25 and printed with 'Tip amount:'.
6. tip is added to running_total using += and the new total is printed.
7. final_bill is calculated by dividing running_total by num_of_friends.
8. each_pays = round(final_bill, 2) is computed and printed as 'Each person pays:'.

Submission (2 marks)

Please submit the following two files to Google Classroom:

- ①

A Python file named: `[YourClass][YourNumber].py` (e.g. `S3D05.py`)
- ②

A screenshot showing your complete code and output in the *Thonny IDE*.

Expected Output

```
Total bill so far: 198.82999999999998
Tip amount: 49.707499999999996
Total with tip: 248.53749999999997
Bill per person: 62.13437499999999
Each person pays: 62.13
```

— — — End of Task — — —

fCC Task: *Workshop: Build a Movie Ticket Booking Calculator*

Skills: *if/elif/else, and, or, not, comparison operators, nested conditionals*

Please use **Thonny** to complete the task. Add these lines at the top of your code:

```
print("Name: [Your Name]")
print("Class: [Your Class]")
print("Class Number: [Your Class Number]")
```

Instructions

Write a Python script that calculates the price of a movie ticket. The price depends on the customer's age (child/adult/senior), whether they have a membership, and whether it is a weekend. Use boolean logic and nested conditionals throughout.

Key Rules & Concepts:

- ▶ and: BOTH conditions must be True for the whole expression to be True.
- ▶ or: AT LEAST ONE condition must be True.
- ▶ not: reverses True to False, and False to True.
- ▶ elif lets you chain multiple conditions — only the FIRST matching branch runs.
- ▶ Nested if = an if statement inside another if statement.

Objective: Follow each step below to complete the task.

Steps to Complete
1. Prompt for age using <code>int(input())</code> and store it in <code>age</code> . Use <code>if/else</code> : <code>if age < 18 → ticket_type = 'child', price = 8.0; else → ticket_type = 'adult', price = 15.0</code> .
2. Set <code>is_senior = age >= 60</code> . If <code>is_senior</code> is <code>True</code> : <code>ticket_type = 'senior', price = 10.0</code> .
3. Prompt for membership: <code>is_member = input('Are you a member? (yes/no): ') == 'yes'</code> .
4. If <code>is_member</code> and <code>ticket_type == 'adult'</code> : apply a \$3.00 discount (<code>price -= 3.0</code>).
5. Prompt for the day. Set <code>is_weekend = day == 'weekend'</code> .
6. If <code>is_weekend</code> and not <code>is_member</code> : apply a \$2.00 surcharge (<code>price += 2.0</code>).
7. Use a nested <code>if</code> : <code>if price > 0</code> : check <code>is_member</code> to decide which message to print.
8. Inside the nested <code>if</code> : <code>if is_member → print 'Member {ticket_type} ticket: \${price}'; else → print '{ticket_type.capitalize()} ticket: \${price}'</code> .

Output Checklist (8 marks)
1. <code>age</code> is collected with <code>int(input())</code> and the child/adult split uses <code>if age < 18</code> .
2. <code>is_senior = age >= 60</code> is used to update <code>ticket_type</code> and price for seniors.
3. <code>is_member</code> is set by comparing the input result to the string <code>'yes'</code> .

4. The membership discount of \$3.00 uses and to check both is_member and ticket_type.
5. is_weekend is a boolean set by comparing the day input to 'weekend'.
6. The weekend surcharge uses and and not correctly.
7. A nested if is used: the outer checks price > 0, the inner checks is_member.
8. The output message correctly uses .capitalize() for non-member tickets.

Submission (2 marks)

Please submit the following two files to Google Classroom:

- ①

A Python file named: [YourClass][YourNumber].py (e.g. S3D05.py)
- ②

A screenshot showing your complete code and output in the *Thonny IDE*.

Sample Runs (three scenarios shown)

```
--- Scenario 1: Adult, member, weekday ---
Enter your age: 25
Are you a member? (yes/no): yes
Day (weekday/weekend): weekday
Member adult ticket: $12.0

--- Scenario 2: Child, no member, weekend ---
Enter your age: 10
Are you a member? (yes/no): no
Day (weekday/weekend): weekend
Child ticket: $8.0

--- Scenario 3: Senior, no member, weekday ---
Enter your age: 65
Are you a member? (yes/no): no
Day (weekday/weekend): weekday
Senior ticket: $12.0
```

— — — End of Task — — —

fCC Task: Lab: Build a Travel Weather Planner

Skills: Multiple variables, if/elif/else chains, comparison operators, boolean expressions

Starter code — copy these variables into your file:

```
# Starter variables — use these exact values
temperature          = 38
humidity              = 80
wind_speed            = 25
is_raining            = False
distance_to_destination = 150
```

Instructions

Write a Python script that gives travel advice based on weather conditions. You are given five starter variables. Write three separate if/elif/else chains to produce advice for temperature, rain, and travel distance.

- Key Rules & Concepts:
- ▶ elif chains: Python checks from top to bottom and runs the FIRST matching branch only.
 - ▶ Boolean variables like is_raining can be used directly: if is_raining: (no == True needed).
 - ▶ You can chain as many elif branches as needed.
 - ▶ Each of the three chains stores its result in a separate variable before printing.

Objective: Follow each step below to complete the task.

Steps to Complete
1. Copy the five starter variables into your script with the exact values shown.
2. Write a temperature chain: if temperature > 35 → temp_advice = 'It is very hot. Stay hydrated!' elif temperature > 20 → 'The weather is nice and warm.' elif temperature > 10 → 'It is a bit chilly. Bring a jacket.' else → 'It is very cold. Dress warmly.'
3. Store the temperature result in a variable named temp_advice.
4. Write a rain chain: if is_raining → rain_advice = 'It is raining. Bring an umbrella!' else → 'No rain expected. Enjoy your trip!'
5. Store the rain result in a variable named rain_advice.
6. Write a distance chain: if distance_to_destination > 100 → travel_tip = 'Long trip ahead. Consider resting stops.' elif distance_to_destination > 50 → 'Medium trip. Pack some snacks.' else → 'Short trip. You will be there soon!'
7. Store the distance result in a variable named travel_tip.
8. Print temp_advice, rain_advice, and travel_tip — one per line.

1. All five starter variables are set with the exact values provided.
2. The temperature chain uses if/elif/elif/else and covers all four temperature ranges.
3. The temperature advice is stored in a variable named temp_advice.
4. The rain chain uses if/else to check is_raining directly (without == True).
5. The rain advice is stored in a variable named rain_advice.
6. The distance chain uses if/elif/else and covers all three distance ranges.
7. The distance advice is stored in a variable named travel_tip.
8. All three advice variables are printed, one per line.

Submission (2 marks)

Please submit the following two files to Google Classroom:

- ① A Python file named: `[YourClass][YourNumber].py` (e.g. `S3D05.py`)
- ② A screenshot showing your complete code and output in the *Thonny IDE*.

Expected Output (using the starter variables)

```
It is very hot. Stay hydrated!
No rain expected. Enjoy your trip!
Long trip ahead. Consider resting stops.
```

— — — End of Task — — —

fCC Task: Lab: Build an Apply Discount Function

Skills: *def, parameters, return, isinstance(), input validation, round()*

Please use **Thonny** to complete the task. Add these lines at the top of your code:

```
print("Name: [Your Name]")
print("Class: [Your Class]")
print("Class Number: [Your Class Number]")
```

Instructions

Write a function named `apply_discount` that takes a price and a discount percentage, validates both inputs, then calculates and returns the final discounted price rounded to 2 decimal places.

Validation & Calculation Rules:

- ▶ price must be a number (int or float). If not, return: 'The price should be a number'
- ▶ discount must be a number (int or float). If not, return: 'The discount should be a number'
- ▶ price must be greater than 0. If not, return: 'The price should be greater than 0'
- ▶ discount must be between 0 and 100 (inclusive). If not, return: 'The discount should be between 0 and 100'
- ▶ Calculate: `final_price = price - (price * discount / 100)`
- ▶ Return: `round(final_price, 2)`

Objective: Follow each step below to complete the task.

Steps to Complete
1. Define a function named <code>apply_discount</code> that takes two parameters: price and discount.
2. Use <code>isinstance(price, (int, float))</code> to check if price is a number. If not, return the string 'The price should be a number'.
3. Use <code>isinstance(discount, (int, float))</code> to check if discount is a number. If not, return the string 'The discount should be a number'.
4. If <code>price <= 0</code> , return the string 'The price should be greater than 0'.
5. If <code>discount < 0</code> or <code>discount > 100</code> , return 'The discount should be between 0 and 100'.
6. Calculate <code>final_price = price - (price * discount / 100)</code> .
7. Return <code>round(final_price, 2)</code> .
8. Test your function by calling it with the values shown in the Tests section.

Tests (8 marks)
1. You should have a function named <code>apply_discount</code> .
2. <code>apply_discount('hello', 10)</code> → should return 'The price should be a number'.

3.	<code>apply_discount(100, 'ten')</code>	→	should return 'The discount should be a number'.
4.	<code>apply_discount(-5, 10)</code>	→	should return 'The price should be greater than 0'.
5.	<code>apply_discount(100, 150)</code>	→	should return 'The discount should be between 0 and 100'.
6.	<code>apply_discount(100, 10)</code>	→	should return 90.0.
7.	<code>apply_discount(200, 25)</code>	→	should return 150.0.
8.	<code>apply_discount(50, 0)</code>	→	should return 50.0.

Submission (2 marks)

Please submit the following two files to Google Classroom:

- ① A Python file named: `[YourClass][YourNumber].py` (e.g. `S3D05.py`)
- ② A screenshot showing your complete code and output in the *Thonny IDE*.

Sample Output

```
The price should be a number
The discount should be a number
The price should be greater than 0
The discount should be between 0 and 100
90.0
150.0
50.0
```

— — — End of Task — — —

ANSWER KEY

Model answers for all 6 fCC tasks. Accept any logically equivalent solution.

Task 1 — Build a Report Card Printer

Chapter: *Ch.2 — Variables & Data Types* | **Skills:** *input(), float(), arithmetic, f-strings with :.1f*

```
print("Name: [Your Name]")
print("Class: [Your Class]")
print("Class Number: [Your Class Number]")

name      = input('Student name: ')
english   = float(input('English grade: '))
maths     = float(input('Maths grade: '))
science   = float(input('Science grade: '))

average   = (english + maths + science) / 3

print('--- Report Card ---')
print(f'Student : {name}')
print(f'English   : {english}')
print(f'Maths      : {maths}')
print(f'Science    : {science}')
print(f'Average   : {average:.1f}')
```

Task 2 — Build an Employee Profile Generator

Chapter: *Ch.3 — Introduction to Strings* | **Skills:** *String concatenation (+), += operator, str() conversion, f-strings, string slicing*

```
print("Name: [Your Name]")
print("Class: [Your Class]")
print("Class Number: [Your Class Number]")

first_name = 'John'
last_name  = 'Doe'
full_name   = first_name + ' ' + last_name

address    = '123 Main Street'
address += ', Apartment 4B'

employee_age = 28
employee_info = full_name + ' is ' + str(employee_age) + ' years old'
print(employee_info)

experience_years = 5
experience_info  = 'Experience: ' + str(experience_years) + ' years'
print(experience_info)

position = 'Data Analyst'
salary   = 75000
employee_card = f'Employee: {full_name} | Age: {employee_age} | Position: {position} | Salary: ${salary}'
print(employee_card)

employee_code = 'DEV-2026-JD-001'
department    = employee_code[0:3]
year_code     = employee_code[4:8]
initials      = employee_code[9:11]
last_three    = employee_code[-3:]
```

```
print(department, year_code, initials, last_three)
```

Task 3 — Build a Bill Splitter

Chapter: *Ch.4 — Numbers & Mathematical Operations* | **Skills:** *Variables, float, += operator, multiplication, division, round()*

```
print("Name: [Your Name]")
print("Class: [Your Class]")
print("Class Number: [Your Class Number]")

running_total = 0
num_of_friends = 4
appetizers = 37.89
main_courses = 57.34
desserts = 39.39
drinks = 64.21

running_total += appetizers + main_courses + desserts + drinks
print('Total bill so far:', running_total)

tip = running_total * 0.25
print('Tip amount:', tip)

running_total += tip
print('Total with tip:', running_total)

final_bill = running_total / num_of_friends
print('Bill per person:', final_bill)

each_pays = round(final_bill, 2)
print('Each person pays:', each_pays)
```

Task 4 — Build a Movie Ticket Booking Calculator

Chapter: *Ch.5 — Booleans & Conditionals* | **Skills:** *if/elif/else, and, or, not, comparison operators, nested conditionals*

```
print("Name: [Your Name]")
print("Class: [Your Class]")
print("Class Number: [Your Class Number]")

age = int(input('Enter your age: '))
if age < 18:
    ticket_type = 'child'
    price = 8.0
else:
    ticket_type = 'adult'
    price = 15.0

is_senior = age >= 60
if is_senior:
    ticket_type = 'senior'
    price = 10.0

is_member = input('Are you a member? (yes/no): ') == 'yes'
if is_member and ticket_type == 'adult':
    price -= 3.0

day = input('Day (weekday/weekend): ')
is_weekend = day == 'weekend'
if is_weekend and not is_member:
```

```
price += 2.0

if price > 0:
    if is_member:
        print(f'Member {ticket_type} ticket: ${price}')
    else:
        print(f'{ticket_type.capitalize()} ticket: ${price}')
else:
    print('Error: invalid price')
```

Task 5 — Build a Travel Weather Planner

Chapter: *Ch.5 Lab — Multiple Conditional Chains* | **Skills:** *Multiple variables, if/elif/else chains, comparison operators, boolean expressions*

```
print("Name: [Your Name]")
print("Class: [Your Class]")
print("Class Number: [Your Class Number]")

temperature          = 38
humidity             = 80
wind_speed           = 25
is_raining            = False
distance_to_destination = 150

if temperature > 35:
    temp_advice = 'It is very hot. Stay hydrated!'
elif temperature > 20:
    temp_advice = 'The weather is nice and warm.'
elif temperature > 10:
    temp_advice = 'It is a bit chilly. Bring a jacket.'
else:
    temp_advice = 'It is very cold. Dress warmly.'

if is_raining:
    rain_advice = 'It is raining. Bring an umbrella!'
else:
    rain_advice = 'No rain expected. Enjoy your trip!'

if distance_to_destination > 100:
    travel_tip = 'Long trip ahead. Consider resting stops.'
elif distance_to_destination > 50:
    travel_tip = 'Medium trip. Pack some snacks.'
else:
    travel_tip = 'Short trip. You will be there soon!'

print(temp_advice)
print(rain_advice)
print(travel_tip)
```

Task 6 — Build an Apply Discount Function

Chapter: *Ch.6 — Understanding Functions & Scope* | **Skills:** *def, parameters, return, isinstance(), input validation, round()*

```
print("Name: [Your Name]")
print("Class: [Your Class]")
print("Class Number: [Your Class Number]")

def apply_discount(price, discount):
    if not isinstance(price, (int, float)):
        return 'The price should be a number'
    if not isinstance(discount, (int, float)):
```

```
        return 'The discount should be a number'
    if price <= 0:
        return 'The price should be greater than 0'
    if discount < 0 or discount > 100:
        return 'The discount should be between 0 and 100'
    final_price = price - (price * discount / 100)
    return round(final_price, 2)
```

```
print(apply_discount('hello', 10))
print(apply_discount(100, 'ten'))
print(apply_discount(-5, 10))
print(apply_discount(100, 150))
print(apply_discount(100, 10))
print(apply_discount(200, 25))
print(apply_discount(50, 0))
```